

Adaptive Handover Optimization in LEO Satellite Networks Using Energy-Aware Q-Learning

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ABSTRACT Low Earth orbit (LEO) satellite networks are rapidly becoming a foundational infrastructure for global connectivity, especially in remote and underserved regions. However, the high mobility of LEO satellites results in frequent handovers, leading to increased signaling overhead, elevated energy consumption and potential service disruptions. To address these challenges, we propose an energy-aware handover management framework based on Q-learning, specifically tailored for single-user, remote scenarios. The proposed scheme dynamically adapts to real-time network conditions, aiming to maximize signal quality while minimizing unnecessary handovers and their energy cost. We evaluate its performance against three baseline methods: traditional Q-learning, predictive modeling, and entropy-based selection. Simulation results show that our approach improves cumulative signal quality by over 50%, reduces handover frequency by up to 65% and achieves higher coverage probability compared to baseline methods, demonstrating its effectiveness in enhancing both service continuity and energy efficiency in LEO satellite communications.

INDEX TERMS Low earth orbit (LEO) satellites, handover optimization, Q-learning, energy-efficiency, signal quality.

I. INTRODUCTION

THE CONVERGENCE of Low Earth orbit (LEO) satellite networks with Internet of Things (IoT) promises global connectivity across diverse application domains such as precision agriculture, wildlife monitoring and disaster response [1], [2]. Unlike terrestrial networks which often struggle to provide reliable coverage in remote or hard-to-reach regions, LEO satellite systems offer scalable and ubiquitous communication infrastructure [3], [4], [5], [6], [7], [8]. This capability is especially valuable for remote use cases such as oil rigs, remote pipelines in deserts and polar regions and maritime operations [9], [10].

However, the high orbital velocity of LEO satellites results in short visibility windows over fixed ground nodes, leading to frequent handovers [11], [12], [13]. These handovers introduce significant signaling overhead, potential service disruptions and critically, elevated energy consumption, posing a major challenge for remote energy-constrained deployments [14], [15], [16]. In such scenarios, sustaining

connectivity while conserving energy becomes essential, particularly when recharging or replacing batteries is impractical.

While numerous handover optimization techniques have been proposed for LEO satellite networks, the majority assume dense user populations and energy abundance. These assumptions do not hold in single-user, remote IoT deployments, where unnecessary handovers can critically deplete limited energy reserves.

To address this challenge, we propose an energy-aware handover optimization framework based on Q-learning. Specifically designed for remote, single-user scenarios, the proposed method dynamically balances signal quality and energy efficiency by incorporating power consumption directly into the learning objective.

A. RELATED WORK

This section reviews prior research efforts aimed at enhancing coverage, improving connectivity and optimizing

resource utilization in LEO satellite networks through handover strategies.

Early handover techniques primarily focused on single-attribute decision criteria, aiming to ensure seamless connectivity while minimizing disruptions in LEO satellite networks. One of the fundamental approaches is Doppler-based positioning, which predicts the optimal handover timing by leveraging satellite motion characteristics [17]. This method utilizes Doppler shift measurements to anticipate the moment when a satellite moves out of range, allowing the network to preemptively establish a connection with a new satellite. Another widely adopted method is predictive modeling, which forecasts satellite movement based on orbital parameters to anticipate handover events before they occur [18]. This approach reduces handover latency by proactively preparing for transitions between satellites.

Channel reservation techniques have also been proposed to dynamically allocate resources based on traffic priorities, reducing handover failure rates and improving overall network performance [19]. These techniques ensure that handovers occur only when sufficient channel resources are available, thereby minimizing the risk of dropped connections. Additionally, graph theory-based methods have been explored for optimizing satellite selection by identifying the shortest path in a dynamically changing network topology [20]. These methods formulate the handover problem as a graph optimization task, enabling efficient route selection while minimizing link disruptions.

Although these methods have proven effective in reducing handover failure rates, they lack adaptability to real-time network conditions. The reliance on precomputed models and static optimization criteria leads to suboptimal performance in highly dynamic environments, where frequent changes in satellite positions and network load demand more flexible and adaptive handover mechanisms. Moreover, these approaches often exhibit higher latency, making them less suitable for modern LEO satellite IoT applications that require real-time connectivity.

More recent research efforts have focused on integrating multi-attribute decision-making and intelligent optimization techniques to enhance the efficiency of LEO satellite handover strategies. One such advancement is the use of time-expanded graph models, which incorporate velocity-aware algorithms to predict optimal handover sequences dynamically [21]. These models consider multiple time instances, enabling more precise handover decisions based on satellite trajectories and user mobility patterns.

Game-theoretic approaches have also been explored, applying potential game theory to balance handover frequency and call-dropping probability [22]. By modeling handover decisions as a strategic interaction between users and satellites, these approaches optimize resource allocation and ensure efficient load distribution across the satellite network.

Furthermore, the adoption of machine learning-based methods has introduced adaptive optimization techniques such as reinforcement learning (RL) and entropy-based approaches, which dynamically refine handover policies based on historical network data and real-time observations [23], [24], [25]. These techniques allow the system to learn from past handover events and adjust decision-making strategies accordingly, improving overall network efficiency.

Most existing handover optimization strategies in LEO satellite networks are designed for multi-user environments, where energy is often treated as an abundant resource [27]. These scenarios are characterized by inter-user interference, dynamic load balancing, and frequent control signalling, all of which contribute to significantly higher energy consumption per device [28], [29]. For instance, multi-user coordination and signalling in LEO systems can increase per-user energy consumption by $\sim 2\text{--}3\times$ due to added processing and control overhead [30], [31].

In contrast, single-user deployments, such as those found in environmental monitoring, polar research and maritime sensing, operate under fundamentally different constraints [9], [10]. These devices are typically isolated, battery powered and unable to benefit from collaborative scheduling or redundancy. Here, energy consumption is primarily driven by link quality and the frequency of handovers, rather than load balancing and inter-user scheduling [32], [33], [34].

Applying conventional strategies in these settings can lead to excessive and unnecessary handovers, accelerating battery depletion and undermining long-term network sustainability [35]. Despite this, the literature continues to emphasize multi-user assumptions, leaving a gap in strategies tailored for energy-constrained, single-user contexts.

This study is motivated by the limited exploration of this scenario in current research. We target the underexplored problem of optimizing handovers for autonomous, energy-constrained single-user devices operating in remote areas. Unlike traditional methods that neglect the impact of frequent handovers on energy reserves, our approach focuses on the development of a lightweight, adaptive and energy-aware handover mechanism that ensures continuous connectivity while extending device operational lifetime in energy-constrained environments.

B. CONTRIBUTION

To address the overlooked energy challenges of single-user, remote deployments of LEO satellite networks, this work proposes a lightweight reinforcement learning (RL)-based handover optimization framework. The main contributions of this paper are as follows:

- 1) *Novel Energy-Aware Reward Function for Q-Learning-Based Handover Optimization:* We introduce a new Q-learning reward formulation that jointly considers signal quality and handover energy cost. This dual-objective design enables the agent to avoid frequent handovers and prioritize sustained, energy-efficient links. Unlike prior works that optimize signal metrics

TABLE 1. A comparison of handover strategies for LEO satellite networks.

Reference	Year	Handover Technique	Optimization Technique
[17]	2004	Uses Doppler effect for predicting handover requests	Doppler-based positioning
[18]	2015	Uses predictive model to anticipate behavior	Predictive modelling
[19]	2015	Reserves channels based on priority	Dynamic threshold adjustment
[20]	2016	Uses graph theory for optimal handover paths	Graph theory
[21]	2018	Considers user velocity for handover	Time expanded graph, shortest path algorithm
[22]	2019	Uses potential game theory for decision making	Game theory
[23]	2020	Agents learn optimal handover strategies	Multi-agent reinforcement learning
[24]	2020	Uses entropy to weigh multiple factors for decision making, multi-objective handover	Entropy method
[25]	2020	Focus on user experience	Reinforcement learning
[26]	2020	Uses predictive model to anticipate behavior	Predictive modelling
This work	2025	Learns energy-efficient handover decisions based on real-time signal quality and energy cost in remote, single-user environments	Energy-aware Q-learning with dynamic reward balancing quality and handover cost.

alone, our method explicitly targets energy preservation, which is critical in battery-powered single-user deployments.

- 2) *Design for Remote, Autonomous Single-User Operation:* While most handover schemes are developed for multi-user networks, our approach operates fully autonomously, without relying on resource coordination, control signaling or infrastructure assistance. This design makes it suitable for isolated deployments such as polar or maritime sensors, scenarios often ignored in existing literature.
- 3) *Comprehensive Comparative Evaluation:* We benchmark our proposed method against energy-unaware Q-learning, entropy-based, and predictive baselines under realistic LEO satellite dynamics. The results show that our strategy significantly improves signal continuity, reduces handover frequency and lowers energy consumption, demonstrating the practical advantage of incorporating energy-awareness into handover policy design.

This single-user focus is particularly important because such deployments face harsher constraints than multi-user systems. A single, remotely deployed IoT device lacks collaborative access scheduling and must autonomously decide when and to which satellite to handover, often under unpredictable link conditions and limited energy reserves. This exposes the device to rapid power depletion if handovers are not carefully optimized. By modeling this isolated scenario, we address a critical but underexplored use case that demands minimal signaling, maximal decision autonomy and robust performance in the absence of external coordination.

II. SYSTEM MODEL

We consider a dynamic LEO satellite communication system consisting of N_s satellites in uniform, circular low Earth orbits. The network is designed to provide continuous coverage to a single, stationary ground-based user located

in a remote area. At any given time, at least one satellite remains within the user's visibility, ensuring uninterrupted connectivity. Due to the rapid orbital motion of LEO satellites, the set of visible satellites changes frequently. Consequently, handover decisions must be made dynamically based on real-time signal quality measurements to maintain reliable service.

A. CHANNEL AND SIGNAL MODEL

The received signal-to-noise ratio (SNR) at time t for a given satellite i is defined as:

$$\gamma_{i,t} = \frac{P_0 |h_{i,t}|^2}{N_0}, \quad (1)$$

where P_0 is the transmit power, $h_{i,t}$ is the channel gain, and N_0 is the noise power. The temporal evolution of the channel gain is modeled as an Ornstein-Uhlenbeck process [36]:

$$h_{t+1,i} = h_{t,i} + \theta(\mu_h - h_{t,i})dt + \sigma_h dB, \quad (2)$$

where θ and represents the mean reversion rate, $\mu_h > 0$ is the long-term mean signal quality, $\sigma_h > 0$ denotes the signal quality volatility and dB represents a Wiener process. μ_h represents the geocentric angle between the user and the satellites, which is connected to the user's elevation angle and can be defined as [37]:

$$\mu_h = h_0 \cos \alpha - \cos \alpha_0 \quad (3)$$

where h_0 denotes the channel gain, and α_0 represents the maximum geocentric angle within the satellite's coverage area.

The value of α_0 is determined by the satellite's orbital height h and the user's minimum visible elevation angle α_e , as follows [37]:

$$\alpha_0 = \sin \left(\arccos \left(\frac{R_e}{R_e + h} \right) \right) \cos \alpha_e - \alpha_e \quad (4)$$

where R_e is the radius of the Earth.

The cumulative received signal quality over a time interval is given by:

$$\int \gamma_{i,t} dt = \frac{P_0}{N_0} \int |h_{i,t}|^2 dt. \quad (5)$$

To optimize connectivity, the handover mechanism seeks to maintain a link with the satellite providing the highest cumulative signal quality. Accordingly, the utility of a handover from satellite i to satellite j at time t is [37]:

$$u(t) = \int_t^{t_j} \gamma_{j,t} dt, \quad (6)$$

where t_j represents the last coverage time of satellite j .

In the energy-aware handover strategy, we incorporate the cost of handover energy consumption into the utility function to strike a balance between service quality and device longevity. The modified utility function becomes:

$$u(t) = \int_t^{t_j} \gamma_{j,t} dt - \lambda E_{\text{handover}}, \quad (7)$$

where E_{handover} denotes the energy consumed during handover and λ is a weighting factor that governs the trade-off between signal quality maximization and energy-efficiency.

Given the remote, energy-constrained nature of the single-user IoT deployment considered in the study, Binary Phase Shift Keying (BPSK) is assumed as the underlying modulation scheme. BPSK is chosen for its high robustness to noise, low complexity, and energy efficiency, making it well-suited for dynamic LEO satellite environments where maintaining reliable connectivity under varying signal conditions is essential.

B. SATELLITE HANDOVER PROCESS

In the proposed scenario, the handover process comprises three main stages: handover information collection, decision-making and execution. The objective is to sustain robust connectivity while minimizing unnecessary handovers and the associated energy expenditure.

1) HANDOVER INFORMATION COLLECTION

The ground user continuously monitors the received SNR from the currently connected satellite. A handover is triggered when the SNR falls below a predefined threshold $\gamma_{\min}$ for a sustained interval, signaling potential link degradation. This real-time monitoring facilitates proactive assessment of alternative satellites before connection quality becomes unacceptable.

2) HANDOVER DECISION-MAKING

The handover decision is formulated as an energy-aware optimization problem. The goal is to maximize cumulative signal quality while minimizing the frequency of handovers and their associated energy costs.

The next satellite $s^* \in J$ is selected by maximizing the utility function over its expected coverage window:

$$s^* = \arg \max_{j \in J} \left(\int_t^{t+T_j} \gamma_{j,t} dt - \lambda E_{\text{handover}} \right), \quad (8)$$

where J denotes the set of candidate satellites, $\gamma_{j,t}$ is the received SNR from satellite j at time t , T_j is the estimated remaining visibility duration of satellite j , E_{handover} is the energy cost of performing a handover and λ is a weight parameter controlling the trade-off between signal quality and energy efficiency.

3) HANDOVER EXECUTION

Once the optimal satellite is selected, the user initiates a seamless soft handover procedure, maintaining simultaneous connections with the current and target satellites during a brief transition phase. This approach minimizes the risk of service interruption.

The probability of handover failure P_f during a session is modeled as [37]:

$$P_f = 1 - (1 - p)^{n_m}, \quad (9)$$

where p is the failure probability of a single handover attempt, and n_m is the total number of handovers executed during the session. To ensure reliability, the following reliability constraints is imposed [37]:

$$P_f \leq \mu, \quad (10)$$

where μ is the maximum allowable handover failure probability. This constraint guarantees that the handover strategy remains both energy efficient and resilient to failure.

C. PROBLEM FORMULATION

The objective of the handover optimization is to determine an optimal policy Ω that maximizes long-term utility by balancing signal quality and energy efficiency. Specifically, the utility function considers the trade-off between maintaining high SNR and minimizing energy cost associated with frequent handovers. The problem is formulated as a long-term average cost minimization problem:

$$\begin{aligned} \min_{\Omega} \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=1}^T (-\gamma_{s,t} + \lambda E_{\text{handover}}) \right], \\ \text{subject to: } P_f = 1 - (1 - p)^{n_m} \leq \mu, \end{aligned}$$

where $\gamma_{s,t}$ is the SNR from the selected satellite s at time t .

Due to the stochastic and partially observable nature of the environment, characterized by dynamic satellite visibility and uncertain link quality, an RL approach is adopted. We employ Q-learning to learn the optimal policy Ω by interacting with the environment and updating action-value estimates based on observed rewards. This allows the agent to converge toward energy-efficient handover decisions that maximize cumulative signal quality while minimizing unnecessary switching costs.

III. REINFORCEMENT LEARNING HANDOVER SCHEME

A. ENERGY-AWARE Q-LEARNING DECISION PROCESS

We adopt an energy-aware Q-learning framework to model the handover decision-making process, wherein the agent (i.e., ground user) interacts with the dynamic LEO satellite environment to learn an optimal policy that balances signal quality, handover frequency and energy consumption.

1) STATE AND ACTION SPACE

At time t , the state s is represented as the vector of received SNR values from all visible satellites:

$$S_t = \{h_{t,s_1}, h_{t,s_2}, \dots, h_{t,s_{N_s}}\},$$

where h_{t,s_i} denotes the signal quality from satellite s_i at time t . The action $a \in A_t$ corresponds to selecting a satellite from the candidate set $J = \{s_1, s_2, \dots, s_{N_s}\}$ for potential handover.

2) REWARD FUNCTION

The reward function incorporates two key components: the received signal quality and the energy cost associated with performing handovers. By explicitly combining these factors, the agent is incentivized to maintain high quality links while avoiding frequent, energy intensive handovers.

$$r = \begin{cases} h_{t,A_t} - \lambda E_{\text{handover}}, & \text{if } h_{t,A_t} > \gamma_{\min} \\ -\lambda E_{\text{handover}}, & \text{otherwise} \end{cases} \quad (11)$$

3) Q-LEARNING UPDATE

The Q-value for a state-action pair is updated using the standard Bellman equation:

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') \right], \quad (12)$$

where α is the learning rate γ is the discount factor, s' is the next state and a' is the next action that maximizes the expected future reward.

4) EXPLORATION-EXPLOITATION STRATEGY

To balance exploration and exploitation, an ϵ -greedy strategy is employed. At each decision step, the agent selects a random action with probability ϵ (exploration), and the action with the highest Q-value with probability $1 - \epsilon$ (exploitation). The exploration rate decays exponentially over time according to $\epsilon_t = \epsilon_0 \cdot (0.999)^t$, where $\epsilon_0 = 0.6$ is the initial exploration rate and t is the episode index. To ensure persistent exploration, ϵ is bounded below by a minimum threshold of 0.1. This decay schedule encourages broad exploration in early training and steady convergence as learning progresses.

B. TRAINING PROCESS AND CONVERGENCE

The training procedure iteratively updates the Q-table across multiple episodes as presented in **Algorithm 1**. At each time step, the agent observes the current, selects an action based on the ϵ -greedy strategy, receives a reward, and updates the Q-value accordingly. This process continues until it

Algorithm 1 Energy-Aware Q-Learning Handover

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1: Initialize Q-table with zeros
2: for each episode do
3:   Observe current state  $s_t$ 
4:   Select action  $a_t$  using  $\epsilon$ -greedy policy
5:   Execute handover (if needed), and observe new state  $s_{t+1}$ 
6:   Measure signal quality and compute energy cost
7:   Compute reward (11)
8:   Update Q-table (12)
9:   Decrease  $\epsilon$  to improve exploitation over time
10: end for
11: Repeat until convergence

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converges to a stable policy that maximizes long-term utility and reduces energy-wasteful handovers.

While our empirical results demonstrate convergence of the proposed approach, it is worth noting that Q-learning is theoretically proven to converge to the optimal action-value function under specific conditions [38], [39]. These include the use of a decaying learning rate (α) that satisfies the Robbins-Monro conditions ($\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$), sufficient exploration of all state-action pairs and a finite Markov decision process (MDP) with bounded rewards. In our implementation, a constant learning rate is used for practical purposes; however, the environment is modeled as a finite MDP with bounded signal quality and energy costs, and exploration is maintained through a bounded ϵ -greedy policy. Although some theoretical assumptions are relaxed, the overall learning setup adheres to key convergence principles established in literature.

For the traditional Q-learning baseline, the reward function was defined to maximize the signal quality without considering energy consumption. Specifically, the agent receives a reward equal to the observed signal quality from the selected satellite if it exceeds a predefined threshold ($\gamma_{\min}$); otherwise, a small penalty is applied. This formulation encourages the agent to select links that provide strong received signal strength but does not penalize frequent switching between satellites. The reward structure was adopted from prior work on signal quality-aware handover using reinforcement learning in LEO satellite networks [37]. This baseline enables a fair comparison with the proposed energy-aware Q-learning (EA-QL) approach, which extends the same core framework by incorporating handover energy cost into the reward to support sustainable decision-making under constraints.

IV. SIMULATION SETUP

To evaluate the performance of the proposed energy-aware Q-learning handover scheme, we conduct extensive simulations in a dynamic LEO satellite network environment. The key simulation parameters are summarized in Table 2.

The effectiveness of the proposed scheme is evaluated using the following key performance indicators;

TABLE 2. Simulation parameters [37].

Parameter	Value
Number of LEO Satellites (N_s)	66
Number of Users	1
Number of Iterations	3000
Learning Rate (α)	0.4
Discount Factor (γ)	0.9
Exploration Rate (ϵ)	0.6
SNR Threshold ($\gamma_{\min}$)	0.5×10^{-8}
Duration of Service	50 s
Energy per Handover (E_{handover})	3 J
Energy cost penalty (λ)	0.2
Satellite Orbit Height (h)	1414×10^3 m
Earth's Radius (R_e)	6378×10^3 m
Satellite Coverage Radius (R_0)	2670×10^3 m
Transmit Power (P_0)	50 W
Noise Power (n_0)	1
Drift Parameter (θ)	1
User visible elevation angle (α_e)	0.001 rad
Volatility of the Channel (σ_h)	2.5×10^{-5}
Failure probability of one handover (p)	0.99

average reward convergence to track the stabilization of the Q-learning algorithm over multiple training episodes, cumulative signal quality, cumulative handovers, coverage probability and energy consumption.

V. RESULTS & DISCUSSION

This section presents the simulation results and evaluates the performance of the proposed energy-aware Q-learning (EA-QL) handover scheme against three baseline strategies based on traditional Q-learning, predictive modeling and entropy-based selection.

To clarify the distinction between the proposed EA-QL and the traditional Q-learning baseline, we note that traditional Q-learning serves as a standard RL benchmark that optimizes satellite selection purely based on maximizing signal quality without energy considerations [37]. In contrast, the EA-QL framework learns to optimize handover decisions by balancing signal strength and energy efficiency in real time, through its composite reward structure. The predictive modeling baseline selects the next satellite by forecasting future visibility based on orbital mechanics and chooses the satellite expected to provide the longest remaining visibility duration [26], while the entropy-based selection technique evaluates multiple metrics, such as current SNR and estimated coverage duration, and assigns entropy-based weights to compute a composite score for satellite selection [24]. These baselines collectively allow us to contrast the proposed energy-aware learning approach against both adaptive and non-adaptive alternatives under LEO satellite conditions.

A. SENSITIVITY ANALYSIS OF λ IN ENERGY-AWARE Q-LEARNING

The sensitivity analysis shown in Fig. 1 clearly illustrates the impact of the energy penalty parameter λ on key performance metrics. As λ increases, average reward and signal quality decline due to the agent's increased reluctance to perform handovers, which are penalized more heavily. Notably, both handover frequency and energy consumption drop sharply between $\lambda = 0$ and $\lambda = 0.2$, highlighting a sensitive operational region where policy behavior shifts significantly. Beyond $\lambda = 0.4$, the marginal benefit of increasing energy penalties diminishes, with minimal additional energy savings but with a continued drop in link quality and utility. This trade-off confirms $\lambda = 0.2$ as an optimal balance point, significantly reducing handovers (from ~ 300 to ~ 120) and energy consumption (from ~ 5.5 J to ~ 2.5 J) while maintaining acceptable signal quality (~ 0.08). These results affirm the effectiveness of incorporating an energy penalty in the reward function to guide sustainable handover behavior under constrained conditions.

B. ENERGY CONSUMPTION

Fig. 2 illustrates the cumulative energy consumption over time for the four handover strategies. The proposed EA-QL consistently exhibits the lowest energy consumption throughout the simulation period. By the end of the episode $t = 3000$ s, EA-QL records a total energy usage of approximately 4.5 J, outperforming traditional Q-learning (6 J), entropy-based handover (14 J) predictive modeling (14 J). This efficiency gain is primarily attributed to the energy-aware reward function employed in EA-QL which explicitly penalizes energy-intensive operations such as frequent handovers and excessive scanning.

C. CUMULATIVE HANDOVER

Fig. 3 illustrates that EA-QL results in the fewest handovers among all strategies. At $t = 3000$, the cumulative number of handovers for EA-QL is significantly lower compared to the other approaches with approximately 750 handovers, while Q-learning, predictive modeling and entropy-based methods exhibit around 2200 and 2800 handovers, respectively. This reduction is attributed to the energy-aware mechanism which penalizes frequent handovers and promotes long-duration and stable connections. In contrast, the baseline methods do not consider energy costs, leading to aggressive handover behavior and increased signaling overhead.

D. CUMULATIVE SIGNAL QUALITY

As illustrated in Fig. 4, the proposed EA-QL scheme achieves the highest cumulative signal quality over the simulation duration, outperforming all baseline methods. At $t = 3000$, EA-QL records a cumulative signal quality of approximately 0.12, compared to 0.06 for traditional Q-learning, 0.035 for the predictive model and 0.022 for the entropy-based approach. This significant improvement

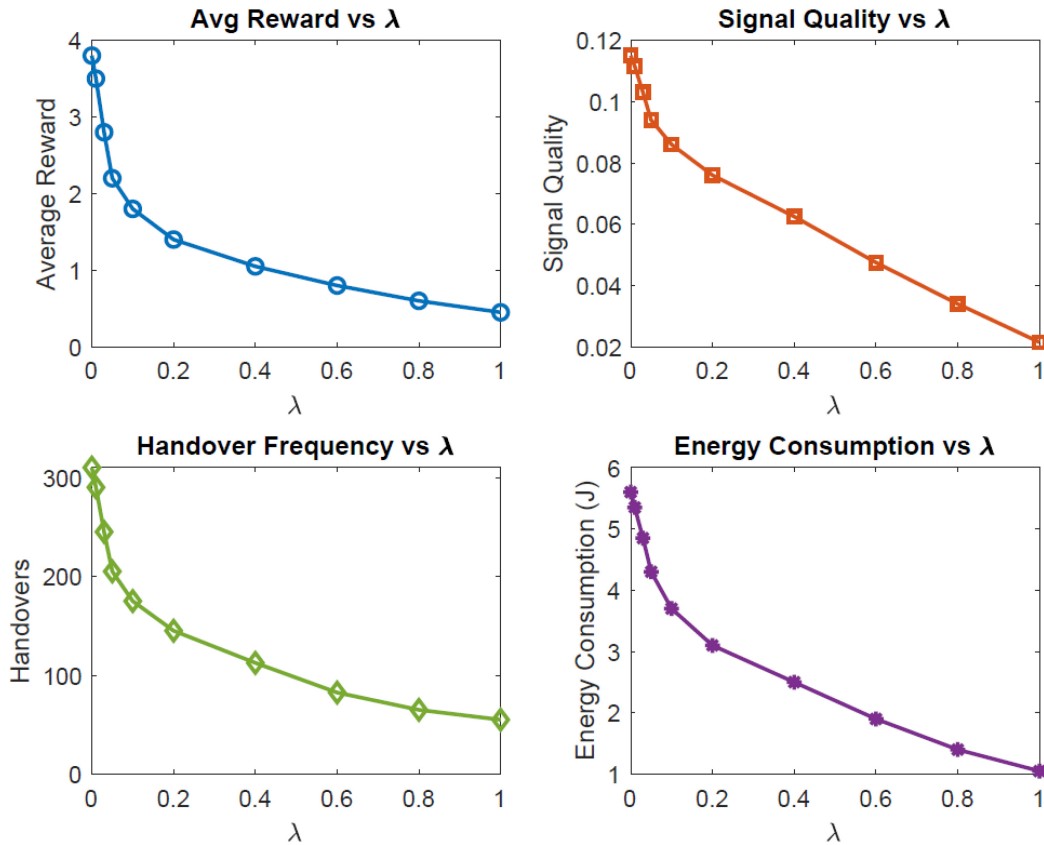


FIGURE 1. Sensitivity analysis of the energy penalty weight λ in the EA-Q Learning reward function.

demonstrates EA-QL's ability to learn and execute handover decisions that prioritize link quality while minimizing unnecessary switching, thereby preserving signal stability over time.

E. COVERAGE PROBABILITY

Fig. 5 depicts the coverage probability attained by each strategy. EA-QL consistently maintains the highest probability of continuous coverage, stabilizing around 0.92. In comparison, conventional Q-learning converges to approximately 0.75, while the predictive and entropy-based approaches settle at 0.58 and 0.40, respectively. The superior performance of EA-QL in maintaining consistent coverage is attributed to its learned policy which favors long-duration connections with strong signal quality, thereby mitigating the risk of coverage loss due to frequent or misinformed handovers.

Traditional Q-learning demonstrates strong performance across all metrics, second only to the proposed EA-Q-learning method. This can be attributed to its ability to adaptively learn a satellite selection policy through direct interaction with dynamic environment. Unlike the predictive modeling and entropy-based baselines, which rely on static heuristics and do not update based on observed outcomes, Q-learning continuously adjusts its policy to maximize signal quality. This feedback-driven learning enables it to outperform fixed-rule strategies, especially in scenarios where

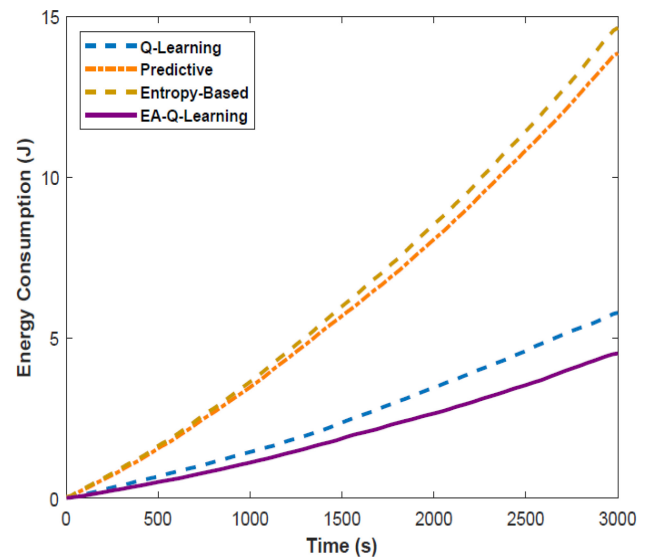


FIGURE 2. Energy consumption over time for different handover strategies.

satellite visibility and signal conditions fluctuate frequently. However, while traditional Q-learning achieves high signal quality, it does not account for energy expenditure and may result in frequent handovers. The EA-QL framework builds on this by incorporating handover energy cost into the reward function, leading to more energy-efficient decision-making while maintaining high link quality.

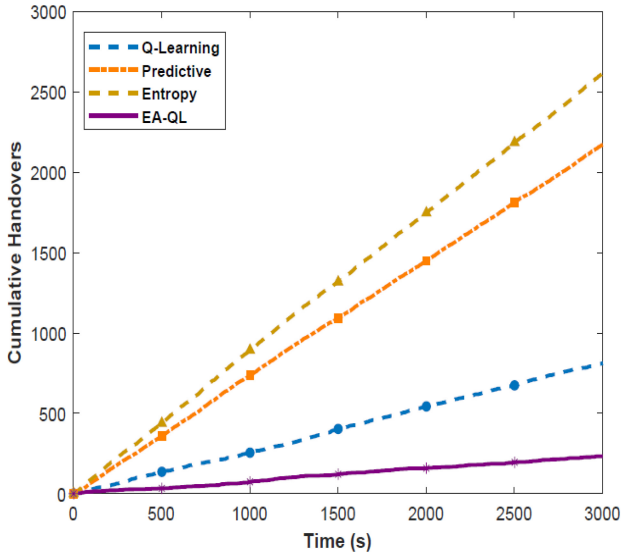


FIGURE 3. Cumulative handovers over time for different handover strategies.

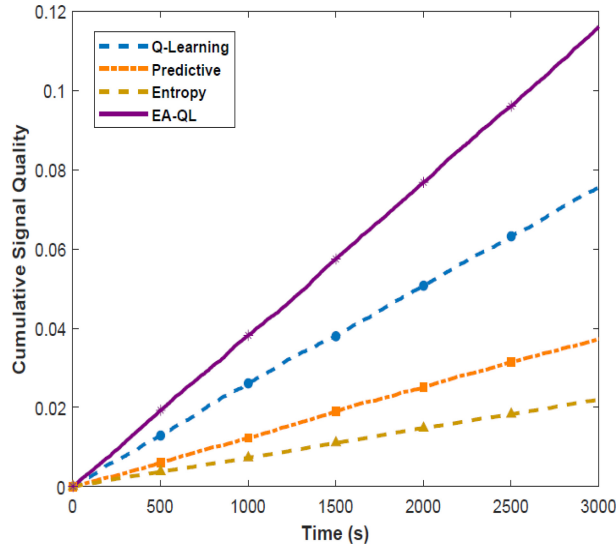


FIGURE 4. Cumulative signal quality over time for different handover strategies.

Although this work primarily focuses on energy-aware handover optimization, we acknowledge that LEO satellite communication inherently faces challenges such as Doppler shifts and signal degradation due to rapid satellite movement and long propagation distances. In our system model, the time-varying nature of the channel, affected by both geometry and orbital dynamics, is captured by modeling signal quality as an Ornstein-Uhlenbeck process. Although our Q-learning framework does not explicitly model or compensate for Doppler effects, its adaptive nature allows the agent to learn policies that implicitly favor more stable links, which are less prone to rapid frequency shifts. Additionally, the use of BPSK modulation provides robustness against moderate Doppler-induced distortion. Future work could incorporate Doppler shift estimation and compensation into the decision process or augment the state space with

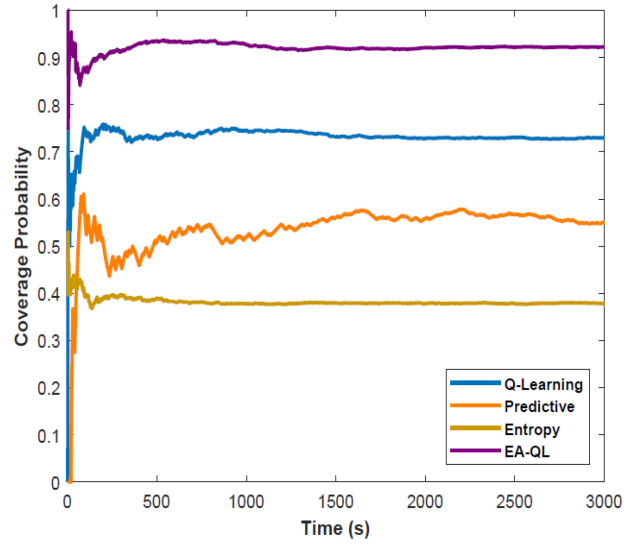


FIGURE 5. Coverage probability over time for different handover strategies.

frequency offset metrics to enhance resilience under extreme mobility conditions.

F. SINGLE-USER VS. MULTI-USER ENERGY CONSUMPTION: THEORETICAL CONSIDERATION

While the proposed EA-QL scheme is tailored for single-user, remote IoT deployments, it is important to consider how energy consumption might behave in multi-user environments. In multi-user scenarios, several devices compete for satellite visibility, increasing the likelihood of overlapping handover attempts and reducing the flexibility to select optimal satellites. This increased contention can lead to suboptimal handovers, more frequent switching and overall higher energy expenditure per user. However, the EA-QL framework's energy-aware reward function is expected to provide some mitigation by discouraging frequent handovers, even in denser settings. Specifically, users independently learn policies that balance link quality with energy cost. This could lead to emergent coordination if future extensions incorporate user-to-user awareness. Although this paper focuses on single-user scenarios, extending EA-QL to multi-user environments and quantitatively analyzing per-user energy consumption under contention is a promising direction for future work.

VI. CONCLUSION

This paper proposed an energy-aware Q-learning-based handover scheme specifically designed for single-user LEO satellite communication networks. The framework addresses the dual challenges of maintaining seamless connectivity and minimizing energy consumption in dynamic, mobility-driven satellite environments. By integrating signal quality and energy cost into the RL reward function, the agent learns to perform intelligent, low-cost handover decisions in real time.

Extensive simulations demonstrate that the proposed energy-aware Q-learning (EA-QL) approach significantly outperforms traditional Q-learning, predictive modeling and entropy-based selection methods. Key findings are that EA-QL achieves the highest cumulative signal quality, it substantially reduces the number of unnecessary handovers, thereby lowering signaling overhead, and it consistently maintains superior coverage probability, resulting in more reliable and energy-efficient connectivity.

While RL provides a powerful tool for adaptive handover management, challenges remain in translating simulation results into practical deployments. Future work will focus on enhancing algorithm scalability, reducing computational complexity for resource-constrained IoT devices and exploring hybrid models that combine RL with predictive analytics to further improve robustness and responsiveness.

Overall, this study lays the groundwork for intelligent and sustainable handover strategies in LEO satellite networks. By leveraging energy-aware RL, future satellite communication systems can achieve massive connectivity with improved service continuity and reduced power consumption which are crucial features for supporting global-scale IoT applications.

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